

OCEANIC GIANT

*This sperm whale has a typically large, elongated head that is continuous with the long, torpedo-shaped body. Like many cetaceans, sperm whales grow to enormous proportions—in males, up to 59 ft (18 m) in length and over 55 tons (50 tonnes) in weight.*

Reproduction

Some whales, including rorquals (such as the humpback whale), breed during the winter. They migrate from their summer feeding grounds in polar seas to tropical waters (usually around island groups or close inshore), where they give birth and then immediately become pregnant again. In spring, they return to colder waters to feed. Other whales may breed seasonally, but do not migrate to do so. In all cetaceans, copulation is very brief. The male's penis, which is S-shaped, is held inside the body wall. It becomes erect not by filling with blood but by straightening as a result of muscular action. After giving birth, the mother (and in some dolphins, other members of the pod) assists the newborn to the surface to take its first breath.

CALF SUCKLING

*Young cetaceans are suckled on milk until they are old enough to eat solid food. The nipples of this Atlantic spotted dolphin are housed in a pouch in the body wall, as is typical of cetaceans.*



CONSERVATION



Modern whaling technology—making hunting more accurate and efficient—threatened certain species with extinction, until the International Whaling Commission (IWC) enforced a moratorium on commercial whaling in 1986. Since then, populations have increased, helped by the setting up of protected areas—such as the 1994 Southern Ocean Whale Sanctuary, which covers the feeding grounds of most of the world's whales. But some species remain “vulnerable,” and pro-whaling nations—such as Japan and Norway—continue to push for a return to commercial quotas.

Surviving underwater

Even though all cetaceans breathe air, they are able to remain underwater for extended periods, returning to the surface only to exhale (“blow”). In order to remain underwater, the heart rate slows by a half. Also, the water pressure squeezes blood out of the vessels near the skin—the blood can then nourish the vital organs. The water pressure compresses the lungs, which forces air into the trachea and nasal passages where some of the air is absorbed by foamy secretions along the respiratory tract wall. Some toothed whales dive to great depths in search of prey.

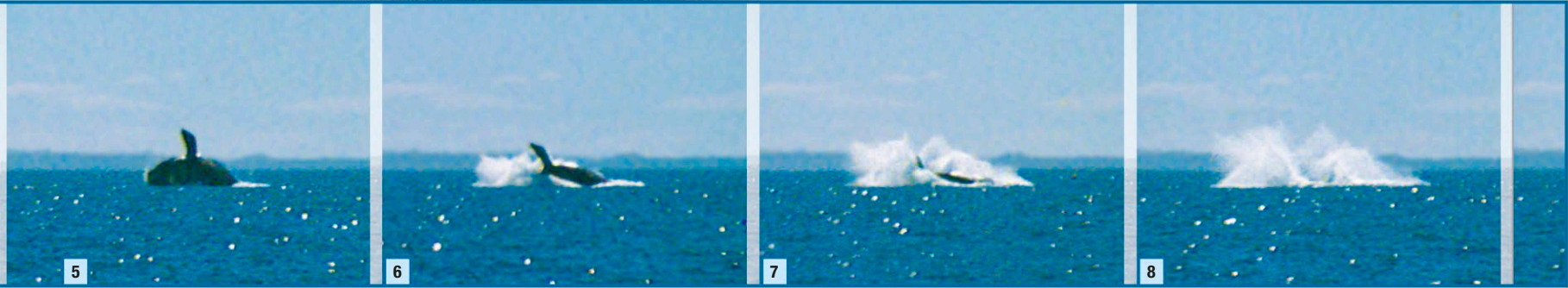
SEARCHING THE DEEP

*Sperm whales in search of squid dive deeper than other cetaceans. They are known to descend to depths of up to 3,300 ft (1,000 m) and can remain underwater for 45 minutes.*



BLOWING

*Cetaceans open their blowholes as they surface and explosively release air and a spray of oil droplets before taking another breath.*



5

REENTERING THE WATER

*The noise made by a whale striking the surface may be heard up to 3,300 ft (1,000 m) away.*

6

GOING UNDER

*As the head disappears, only one front flipper and part of the body is visible.*

7

MAKING A SPLASH

*The whale's reentry creates a huge splash compared with the initial breach of the water's surface.*

8

THE WHALE DISAPPEARS

*With the sequence complete, the whale may breach again. Why whales breach is not clearly understood.*